ENGINEERING RESEARCH REPORT

NACA 2412 Airfoil Aerodynamic Characterization

Low-Reynolds Numerical Simulation for Fixed-Wing Small Unmanned Aerial Vehicles

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1. TITLE PAGE & ABSTRACT

Abstract

This engineering research report presents a numerical aerodynamic analysis of the asymmetric NACA 2412 airfoil operating at a low chord Reynolds number of $Re = 200{,}000$. Utilizing the XFLR5 (v6.62) environment powered by the XFoil viscous panel algorithm, an angle-of-attack (α) sweep was executed across 17 discrete calculation points ranging from $-4.00°$ to $+12.00°$ in steps of $1.00°$. Key evaluated parameters include lift (C_L), drag (C_D), pressure drag (C_{Dp}), pitching moment (C_m), minimum pressure coefficient ($C_{p,min}$), center of pressure location (X_{CP}), and upper/lower boundary layer transition points (X_{tr}). The analysis established a maximum aerodynamic efficiency ratio of $(C_L/C_D)_{max} = 67.31$ at $\alpha = 6.00°$. The findings provide quantitative aerodynamic data tailored for the preliminary structural and flight-dynamics design of small-scale fixed-wing UAVs.

Document Metadata & Executive Summary

- **Airfoil Profile:** NACA 2412 (4-Digit Series Cambered Profile)
- **Numerical Tool:** XFLR5 Direct Foil Analysis (v6.62)
- **Operating Sweep Range:** $\alpha = -4.00°$ to $+12.00°$ ($\Delta\alpha = 1.00°$, 17 calculation points)
- **Design Reynolds Number:** $Re = 200{,}000$
- **Critical Transition Factor:** $N_{crit} = 9.0$ (Standard low-turbulence ambient environment)

2. INTRODUCTION & RESEARCH OBJECTIVES

Small unmanned aerial vehicles (UAVs) operating at low airspeeds encounter low Reynolds number flight regimes ($Re < 500{,}000$). In this domain, boundary layer behavior is highly susceptible to laminar separation bubbles, early transition to turbulence, and adverse pressure gradients, which significantly influence lift-to-drag performance and longitudinal stability.

Primary Objectives:

- Evaluate the sectional lift, drag, and moment characteristics of the NACA 2412 airfoil under steady incompressible flow at $Re = 200{,}000$.
- Identify the optimum angle of attack (α_{opt}) achieving maximum aerodynamic endurance (C_L/C_D ratio).
- Track the progression of upper-surface boundary layer transition ($X_{tr,top}$) and its correlation with suction peak intensification ($C_{p,min}$).
- Provide a baseline aerodynamic dataset for subsequent 3D wing planform synthesis and equilibrium trim analysis.

3. NACA 2412 GEOMETRY & AERODYNAMIC THEORY

The NACA 2412 profile is defined by three main geometric parameters according to standard 4-digit nomenclature:

- **Digit 1 (2):** Maximum camber is **2%** of the chord ($0.02c$).
- **Digit 2 (4):** Location of maximum camber is at **40%** of the chord from the leading edge ($0.40c$).
- **Digits 3–4 (12):** Maximum profile thickness is **12%** of the chord ($0.12c$).

> **Governing Sectional Aerodynamic Coefficients:**
>
> $C_L = L / (q_\infty c)$ | $C_D = D / (q_\infty c)$ | $C_m = M_{c/4} / (q_\infty c^2)$ | $q_\infty = \frac{1}{2} \rho V_\infty^2$

4. METHODOLOGY & XFLR5 SIMULATION SETUP

The numerical computations were conducted in XFLR5 using the interactive **Direct Foil Analysis** module. The solver couples a two-dimensional panel method for potential flow with an interactive boundary layer formulation (e^N envelope method) to account for viscous friction and boundary layer transition.

Fluid & Boundary Conditions

- **Chord Length (c):** 1.000 m
- **Reynolds Number:** 200,000 (Fixed Speed)
- **Mach Number:** 0.000 (Incompressible)
- **Fluid Density (ρ):** 1.225 kg/m³
- **Kinematic Viscosity (ν):** 1.50×10^{-5} m²/s
- **Critical Factor (N_{crit}):** 9.000

Direct Foil Analysis Control Parameters

- **Analysis Sweep:** Sequence Angle of Attack (α)
- **Sweep Range:** Start = -4.00°, End = 12.00°
- **Step Increment ($\Delta\alpha$):** 1.00°
- **Boundary Layer Initialization:** Init BL Enabled
- **Operating Point Storage:** Store Opp Enabled

Numerical Convergence & Transition Modeling

Boundary layer transition is evaluated via the e^N method, where amplification of Tollmien-Schlichting waves reaches critical instability at $N = N_{crit} = 9.0$. Forced transitions were set at $X_{tr,top} = 1.00$ and $X_{tr,bot} = 1.00$ to allow natural free transition modeling along both upper and lower surface contours.

5. COMPLETE NUMERICAL DATASET (17 CALCULATION POINTS)

The table below provides all 17 discrete calculation points extracted directly from the XFLR5 simulation run for NACA 2412 at $Re = 200{,}000$.

α (°)	C_L	C_D	$C_{D,p}$	C_m	C_L / C_D	$C_{p,min}$	X_{CP}	Top Trans.	Bot Trans.
-4.00	-0.28596	0.01487	0.00732	-0.04298	-19.23	-1.8102	0.0920	0.9677	0.1711
-3.00	-0.13036	0.01307	0.00621	-0.05271	-9.97	-1.2681	-0.1656	0.9465	0.3339
-2.00	0.00115	0.01116	0.00552	-0.05613	0.10	-0.8616	49.9819	0.9169	0.6260
-1.00	0.11535	0.01010	0.00515	-0.05259	11.42	-0.5646	0.7087	0.8793	0.8768
0.00	0.28019	0.01000	0.00483	-0.06038	28.02	-0.5873	0.4644	0.8302	0.9795
1.00	0.42187	0.00982	0.00423	-0.06633	42.96	-0.7103	0.4049	0.7618	1.0000
2.00	0.51592	0.01010	0.00412	-0.06251	51.08	-0.8349	0.3679	0.6901	1.0000
3.00	0.61110	0.01072	0.00435	-0.05883	57.01	-1.0266	0.3421	0.6238	1.0000
4.00	0.70557	0.01144	0.00490	-0.05504	61.68	-1.3126	0.3229	0.5592	1.0000
5.00	0.79964	0.01226	0.00555	-0.05122	65.22	-1.6873	0.3080	0.4898	1.0000
6.00	**0.89188**	**0.01325**	**0.00643**	**-0.04729**	**67.31**	**-2.1433**	**0.2959**	**0.4101**	**1.0000**
7.00	0.97699	0.01476	0.00760	-0.04258	66.19	-2.6434	0.2854	0.3002	1.0000
8.00	1.04627	0.01777	0.00994	-0.03627	58.88	-3.1689	0.2752	0.1612	1.0000
9.00	1.10549	0.02165	0.01348	-0.02881	51.06	-3.6974	0.2654	0.0930	1.0000
10.00	1.15277	0.02594	0.01785	-0.01996	44.44	-4.1797	0.2553	0.0683	1.0000
11.00	1.19940	0.03101	0.02313	-0.01238	38.68	-4.6491	0.2469	0.0552	1.0000
12.00	1.24291	0.03719	0.02956	-0.00618	33.42	-5.1293	0.2400	0.0459	1.0000

6. GRAPHICAL POLARS & AERODYNAMIC ANALYSIS

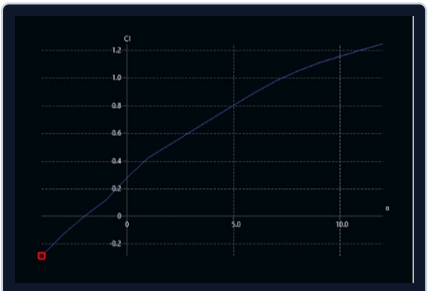


Fig 1: C_L vs. α Lift Curve

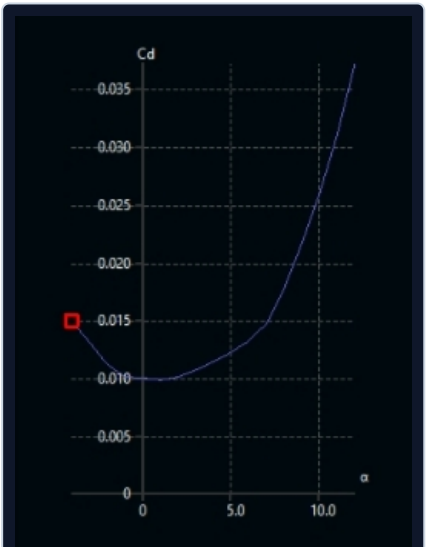


Fig 2: C_D vs. α Drag Curve

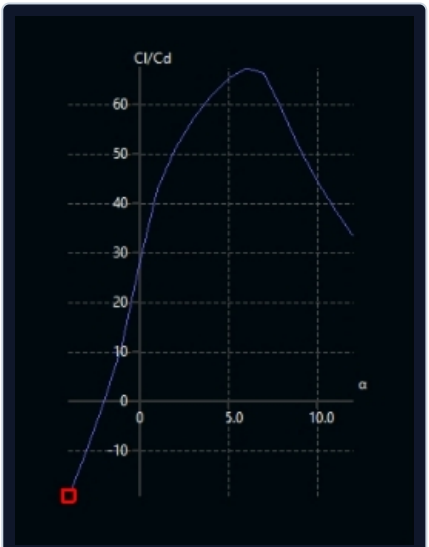


Fig 3: Efficiency C_L/C_D vs α

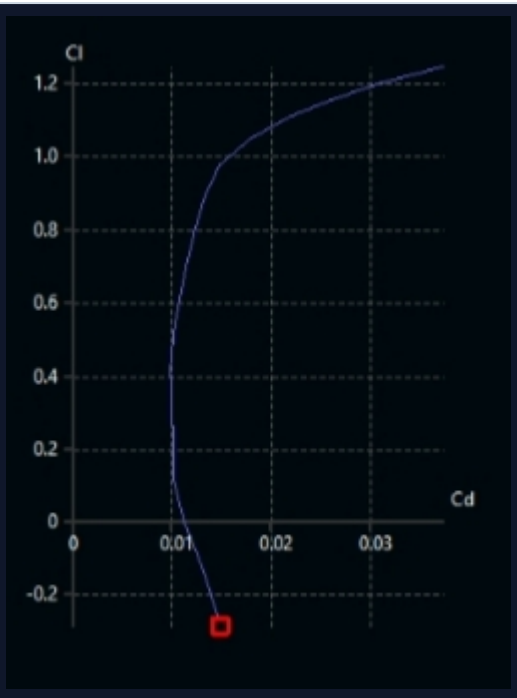


Fig 4: Drag Polar (C_L vs C_D)

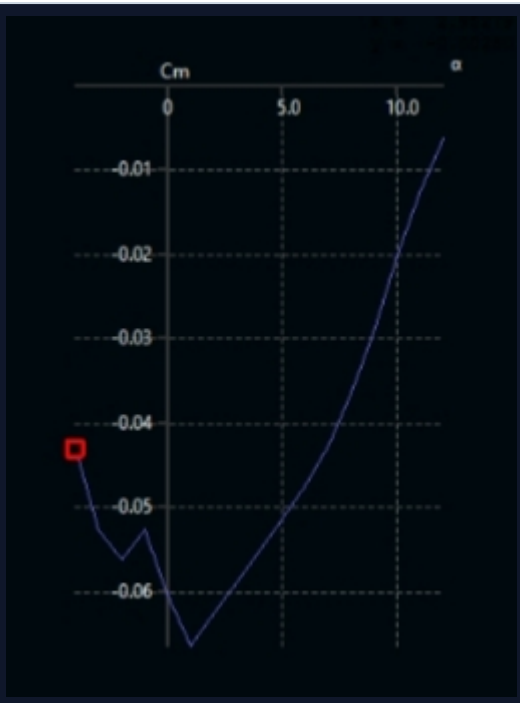


Fig 5: Pitching Moment C_m vs α

Detailed Explanation of Plotted Characteristics

- **Lift Slope (Fig 1):** Demonstrates near-linear growth from $\alpha = -2°$ ($C_L \approx 0$) to $\alpha = 7°$, with the lift curve slope $C_{L,\alpha} \approx 0.098\ deg^{-1}$.
- **Drag Bucket & Polar (Fig 2 & 4):** Minimum drag $C_{D,min} = 0.00982$ is recorded at $\alpha = 1.00°$. Pressure drag (C_{Dp}) accounts for 43% of total drag at $\alpha = 1°$, growing to 79% at $\alpha = 12°$ due to pressure separation.
- **Aerodynamic Ratio (Fig 3):** Peaks sharply at $\alpha = 6.00°$ with $(C_L/C_D)_{max} = 67.31$, representing the optimal angle of attack for maximum UAV flight range.
- **Pitching Moment Behavior (Fig 5):** C_m remains negative across all positive angles, indicating a nose-down pitching tendency characteristic of cambered airfoils.

7. DISCUSSION — INTERPRETATION OF RESULTS

The numerical output highlights critical aerodynamic phenomena pertinent to low-Reynolds UAV design:

- **Boundary Layer Transition Mechanics:** On the upper surface, boundary layer transition moves forward dramatically from $X_{tr,top} = 0.7618$ at $\alpha = 1°$ to $X_{tr,top} = 0.0459$ at $\alpha = 12°$. This is directly driven by the suction peak growth ($C_{p,min}$ dropping from -0.71 to -5.13).
- **Center of Pressure Migration:** As α increases from $0°$ to $12°$, X_{CP} shifts steadily forward from **0.4644c** to **0.2400c**. This forward movement reduces static margin, requiring careful horizontal tail sizing for UAV equilibrium trim.

8. CONCLUSION, LIMITATIONS & FUTURE WORK

Key Conclusions

1. The NACA 2412 delivers high aerodynamic efficiency at low Reynolds numbers, achieving $C_L/C_D = 67.31$ at $\alpha = 6.00°$.
2. The profile exhibits steady lift generation up to $C_L = 1.2429$ at $\alpha = 12.00°$ without abrupt early stall.
3. Pitching moment remains stable, providing predictable longitudinal dynamics for UAV flight control.

Simulation Limitations & Recommended Extensions

- **2D Sectional Limitations:** XFLR5 2D panel analysis ignores 3D wing-tip vortex effects, induced drag (C_{Di}), and fuselage interaction.
- **Post-Stall Behavior:** The viscous-inviscid panel solver does not resolve deep stall hysteresis or massive unattached separated flows above $\alpha > 12°–14°$.
- **Future Improvements:** Validate results using 3D Vortex Lattice Method (VLM2 / 3D Panel) in XFLR5 and high-fidelity RANS CFD solvers, followed by wind-tunnel experimental testing.